Lab 09

PH142 Summer 2026

Announcements

- **8/5: Quiz21, Lab09**
- **8/6: Quiz22**
- **8/7: Quiz23**

Data Project Part III is due next Friday. Start working on it early!

Lab09 Lecture Review

One Sample Proportion Test

Goal: To test if a population proportion equals a claimed value

Assumptions:

- SRS with independent observations
- Sampling distribution is normal ($np_0 \geq 10 \text{ and } n(1 - p_0) \geq 10$)

Lab09 Lecture Review

One Sample Proportion Test

Hypotheses:

$$H_0 : p = p_0$$
$$H_a : p \neq p_0 \ (\text{or} < \text{or} >)$$

Test Statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Lab09 Lecture Review

One Sample Proportion Test

Large Sample Method: $\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Plus Four Method: $\tilde{p} = \frac{\text{number of successes} + 2}{n+4}$

$$\tilde{p} \pm z^* \sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}}$$

Lab09 Lecture Review

Two Sample Proportion Test

Goal: To test if two population proportions are the same or different.

Assumptions:

- SRS from each independent population
- Sampling distribution is normal (observed successes and failures are > 10 for both samples)

Lab09 Lecture Review

Two Sample Proportion Test

Hypotheses:

$$H_0 : p_1 - p_2 = 0$$
$$H_a : p_1 - p_2 \neq 0 \text{ (or } < \text{ or } > \text{)}$$

Test Statistic:

$$z = \frac{\hat{p_1} - \hat{p_2}}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Lab09 Lecture Review

Two Sample Proportion Test

Large Sample Method: $(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

Plus Four Method:

$$\tilde{p}_1 = \frac{\text{no. of successes in pop1} + 1}{n_1+2}$$

$$\tilde{p}_2 = \frac{\text{no. of successes in pop2} + 1}{n_2+2}$$

$$(\tilde{p}_1 - \tilde{p}_2) \pm z^* \sqrt{\frac{\tilde{p}_1(1-\tilde{p}_1)}{n_1+2} + \frac{\tilde{p}_2(1-\tilde{p}_2)}{n_2+2}}$$

LAB 09 Walkthrough

Lab Submission

- Follow the directions on the LAB09 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
  - Copy and paste the given line into the terminal
  - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- **<u>CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED</u>**